Week 20 Homework: TRIZ, Systems, and Models

Answer in your own words. Clear examples are more important than technical vocabulary.

Student: _____ **Date:** _____

Directions. Write about 100–150 words for each answer. You may use a smart-grid example or another familiar example. No coding is required.

1. **What is TRIZ?** Explain what TRIZ is for. Then describe one idea from TRIZ—for example, finding a contradiction, using available resources, dividing a problem into parts, preparing beforehand, or adding feedback—that could help someone improve a research project.

2. **What is a system, based on your understanding?** Give one example. Identify the system's purpose, boundary, important parts, inputs, interactions, and outputs. Briefly explain why changing one part might affect another part.

3. **What is a model, based on your understanding?** Explain how a model differs from the real system it represents. State what your example model includes, what it leaves out, and what evidence could show whether it is useful for a particular question.

Final check: I used my own explanation, gave a specific example, and stated at least one limitation or boundary where appropriate.